Tejaswini Katale

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Education

Masters in Computer Science — Aug 2025 – May 2027
University of Houston, Texas — GPA: 3.7/4.0

Bachelors of Technology in Computer Engineering — Aug 2020 – May 2024
Vishwakarma Institute of Technology, India — GPA: 3.6/4.0

Selected Publications

- **LLM-Guided Hybrid Simulation for Airport Cyber-Resilience Assessment** Link
  Tejaswini Katale, Lu Gao, Yongxin Liu, Dahai Liu, Hongyun Chen
  Mathematics, Special Issue Mathematical Methods in System Engineering Modeling and Simulation, 2026.
- **Developing an AI Assistant for Knowledge Management and Workforce Training in State DOTs** Link
  Divija Amaram, Lu Gao, Gowtham Reddy Gudla, Tejaswini Katale
  Electronics, 2026.
- **Large Language Models in Pavement Safety Analysis: A Comparative Study of Open-Weight Models**
  Tejaswini Katale, Gowtham Reddy Gudla, Lu Gao, Nasir Gharaibeh
  Accepted, ASCE Transportation Conference, 2026
- **A Bilevel Optimization Framework for Adversarial Control of Gas Pipeline Operations** Link
  Tejaswini Katale, Lu Gao, Yunpeng Zhang, Alaa Senouci
  Actuators, 2026.
- **Exploring OSHA Construction Accident Records with Multimodal AI Models**
  Islem Sahraoui, Tejaswini Katale, Gowtham Reddy Gudla, Lu Gao
  Accepted, CI & CRC Joint Conference, 2026
- **Integrating pavement condition records with LLM-based crash narrative analysis for pavement safety assessment** Link
  Sarayu Varma Gottimukkala, Lu Gao, Nasir Gharaibeh, Yussuf Tarnini, Mohammad Talebzadeh, Boni Kutela, Tejaswini Katale
  Accident Analysis & Prevention, Elsevier, 2026.
- **Recession Forecasting: Domain Adapted Feature Selection Technique** Link
  Tejaswini Katale, Jayashri Hase, Aakash Chotrani, Sakshi Ozarde
  InCoWoCo, IEEE, 2024. **Best Paper Award**.

Experience

Research Assistant
University of Houston, Prof. Lu Gao — *Aug 2025 – Present*

- Designed end-to-end LLM evaluation workflows for transportation document retrieval and crash-data analysis tasks.
- Processed 25k+ safety narratives and 600+ DOT documents for large-scale LLM and RAG benchmarking studies.
- Calibrated airport cyber-resilience simulations to measure passenger flow, queue dynamics, and operational impacts.
- Built graph-based forecasting pipelines using roadway topology, traffic patterns, and crash-risk labels.
- Automated SUMO simulation experiments to evaluate adaptive traffic signal control under partial vehicle detection.

Graduate Research Assistant
R.H. Sapat College of Engineering, India — *Sep 2024 – Jun 2025*

- Improved classification accuracy to 97% on 200k+ imbalanced records using a statistical up-sampling framework.
- Outperformed K-means by 16% across five datasets using a skewness-aware clustering approach.
- Reduced feature redundancy across eight datasets while maintaining recall variation below 0.5%.

Research Intern
Vishwakarma Institute of Technology, India — *Dec 2023 – May 2024*

- Achieved 95% accuracy across seven ML models using feature selection on more than 45 features.
- Outperformed K-means and DBSCAN with Python-based clustering methods, achieving 88% accuracy.

Research Projects

LLM-Guided Hybrid Simulation for Airport Cyber-Resilience Assessment

- Developed a hybrid framework combining discrete-event simulation, JuPedSim pedestrian modeling, and structured LLM-guided passenger decision modeling for 500 passengers under cyber disruption.
- Modeled human behavior under delayed or unreliable airport information, including waiting, seeking staff help, checking displays, gate-directed movement, and wrong-direction movement.
- Analyzed 1 baseline and 4 disruption scenarios, showing throughput drop from 21.68 to 11.31 passengers/hour and clearance time rising to 44.21 h.

WasteLess: Mobile App for Reducing Food Waste

- Built a cross-platform mobile application to connect people with extra leftover food to individuals who needed it, with the goal of reducing food waste and improving access.
- Designed the project around everyday user behavior and decision-making, focusing on how people share available food quickly and respond to community needs in real-world settings.

Open-Weight LLMs for Pavement Safety Analysis

- Benchmarked 9 open-weight LLMs ranging from 2B–8B parameters on 25k+ crash narratives for safety classification.
- Achieved F1 scores up to 0.98, supporting comparative analysis of domain-specific LLM performance for pavement safety.

AI Assistant for Knowledge Management and Workforce Training in State DOTs

- Developed a multi-agent RAG assistant over 521 State DOT technical documents, integrating text and figure-based retrieval for pavement knowledge management.
- Evaluated the system on 100 domain-specific queries, achieving Precision@3 of 1.0 and Recall@3 of 94.4%, outperforming single-pass RAG Recall@5 of 58%.

Spatio-Temporal Graph Learning for Crash Risk Forecasting

- Designed an upstream-aware spatio-temporal graph learning model for directional crash risk forecasting using roadway and traffic data.
- Modeled spatial dependency and temporal traffic patterns to support proactive crash-risk prediction and transportation safety analysis.

Honors & Achievements

NSF-supported CyberSTAR Workshop: CyberTraining for Secure Transportation *June 2026*
Selected participant and awarded a $500 stipend

Second Cyber Care Annual Symposium *April 2026*
Embry-Riddle Aeronautical University; presented research project, *LLM-Guided Hybrid Simulation for Airport Cyber-Resilience Assessment*

Coogs for Energy Hackathon *April 2026*
Won 1st place and received a $6,000 prize among 50 teams

Academic Service

Student Committee Member *May 2026*
USDOT Cyber Care UTC National Transportation Cybersecurity Competition

Reviewer *2025 – 2026*
IEEE Conferences: ICDSIS-2025, ICICKE-2025, ICDCECE-2026